sa400_led_monitor v0.2.4 — SA400 form-factor drive-select / write-gate LED monitor

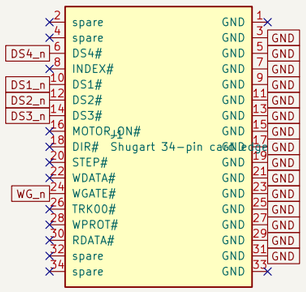
Shugart bus signals are active-low, open-collector, terminated (150R) at the LAST DRIVE. This board is a high-impedance tap: 74LS14 inputs plus 10k pull-ups only.

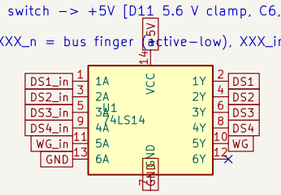
U1 74LS14 inverts the bus: DSn and WG are HIGH when the line is asserted. U3 74LS06 (open collector) sinks the RED LED when DSn is high. U2 74LS00 sinks the GREEN LED when DSn AND WG are both high.

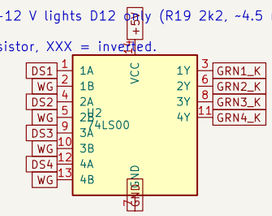
LED current = (5 − 0.4 − Vf) / 470R ~ 5.5 mA with a 2.0 V standard LED (within the 74LS00 8 mA sink). High-efficiency LEDs: use 1k5 (~1.8 mA).

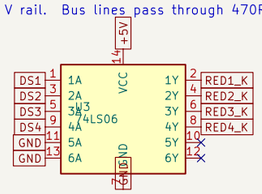
Power (as the SA400): J2-4 +5 V -> F1 PTC -> Q1 P-MOSFET reverse switch -> +5V [D11 5.6 V clamp, C6, C4] -> logic. J2-1 +12 V lights D12 only (R19 2k2, ~4.5 mA). D13 shows the +5 V rail. Bus lines pass through 470R (R14-R18) before the pull-ups and 74LS14.

Unused gate inputs tied to GND; unused outputs left open. Net names: XXX_n = bus finger (active-low), XXX_in = after the series resistor, XXX = inverted.

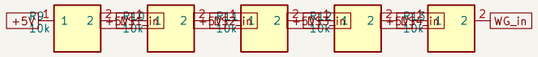


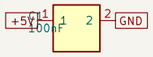


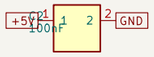


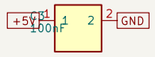


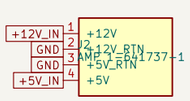


















LED chains: +5V — R — LED anode ; LED cathode — gate output (sinks when active)

























Generated by gen/build.py
RED_n = DSn active. GREEN_n = DSn AND WRITE GATE. All bus signals active-low.
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File: sa400_led_monitor.kicad_sch

Title: SA400 form-factor drive-select / write-gate LED monitor

Size: A3	Date: 2026-10-19	**Rev: v0.2.4**
KiCad E.D.A. 10.0.6		Id: 1/1